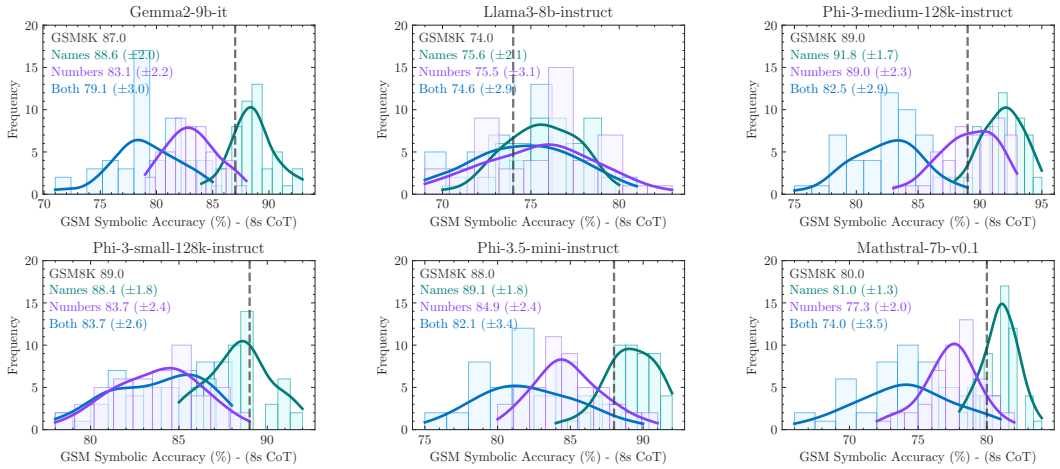

Figure 4: How sensitive are LLMs when we change only names, only proper numbers, or both names and numbers? Overall, models have noticeable performance variation even if we only change names, but even more when we change numbers or combine these changes.

4.2 How Fragile is Mathematical Reasoning in Large Language Models?

In the previous sub-section, we observed high performance variation across different sets generated from the same templates, along with a performance degradation compared to the original GSM8K accuracy. This suggests that the perceived reasoning process of language models may not be formal and is hence susceptible to changes. One explanation is that these models attempt to perform a kind of in-distribution pattern-matching, aligning given questions and solution steps with similar ones seen in the training data. As no formal reasoning is involved in this process, it could lead to high variance across different instances of the same question. In this sub-section and the next one, we investigate these observations further and we show that several factors contribute to the performance variation of the models. First, we investigate the impact of the *type* of change to understand the difference between changing names (e.g., person names, places, foods, currencies, etc.) versus changing numbers (i.e., the values of variables).

Figure 4 demonstrates that while performance variation persists, the variance is lower when changing names compared to numbers. Notably, the original `GSM8K` accuracy of models is now much closer to the center of the changed proper names distribution, in contrast to changed numbers or both. Furthermore, a gradual shift in the means of distributions from right to left, along with an increase in variance, is evident across almost all models. It is both striking and concerning that such performance variance exists when only changing proper names, as this level of variability would not be expected from a grade-school student with genuine mathematical understanding.

From the results in this section, we observe that by increasing the *difficulty* of changes (from names to numbers), the performance drops and the variance increases, overall suggesting that the reasoning capabilities of state-of-the-art LLMs are fragile for the aforementioned reasons. Assuming that LLMs are not performing formal reasoning, how important is the question *difficulty* on the distribution of performance? In the next section, we study this question further.

4.3 How Does Question Difficulty Affect Performance Distribution?

The results in the previous subsection motivate us to study the impact of question *difficulty* on the mean and variance of the performance distribution. To this end, we generate several new templates from the `GSM-Symb`, as illustrated in Fig. 5. First, by removing one clause, we obtain `GSM-Symbolic-Minus-1` or `GSM-M1` for short. Similarly, we can add one or two clauses

Different Levels of GSM-Symbolic Difficulty

GSM-Symbolic-M1: To make a call from a phone booth, you must pay $0.6 for each minute of your call. ~~After 10 minutes, that price drops to $0.5 per minute~~.How much would a 60-minute call cost?

GSM-Symbolic: To make a call from a phone booth, you must pay $0.6 for each minute of your call. After 10 minutes, that price drops to $0.5 per minute. How much would a 60-minute call cost?

GSM-Symbolic-P1: To make a call from a hotel room phone, you must pay $0.6 for each minute of your call.After 10 minutes, that price drops to $0.5 per minute. After 25 minutes from the start of the call, the price drops even more to $0.3 per minute.How much would a 60-minute call cost?

GSM-Symbolic-P2: To make a call from a hotel room phone, you must pay $0.6 for each minute of your call. After 10 minutes, the price drops to $0.5 per minute. After 25 minutes from the start of the call, the price drops even more to $0.3 per minute. If your total bill is more than $10, you get a 25% discount. How much would a 60-minute call cost?

Figure 5: Modifying the difficulty level of GSM-Symbolic by modifying the number of clauses.

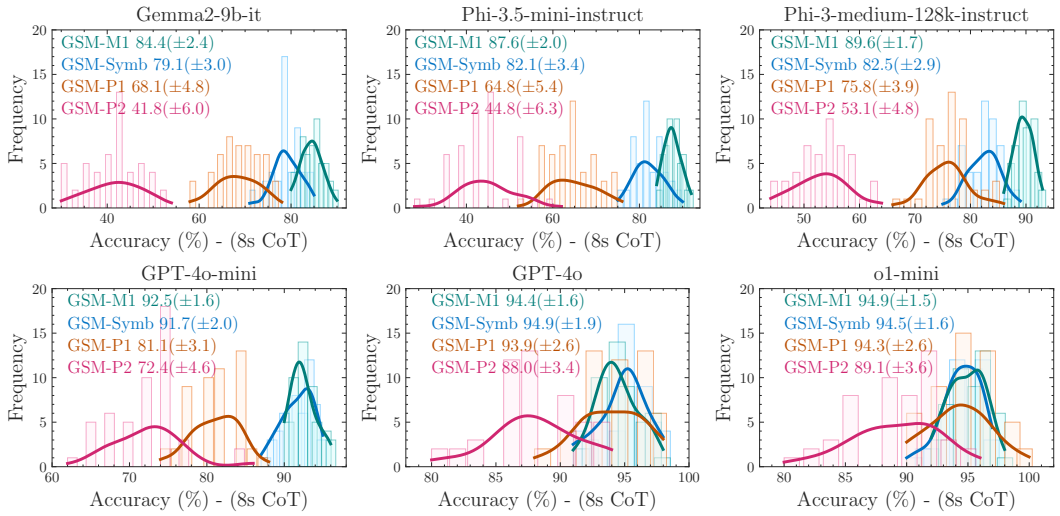

Figure 6: The impact of increasing the number of clauses on performance: As the difficulty increases from GSM-M1→ GSM-Symb→ GSM-P1→ GSM-P2, the distribution of performance shifts to the left (i.e., accuracy decreases), and the variance increases.

to the questions to increase the difficulty, resulting in GSM-Symbolic-Plus-1 (GSM-P1) and GSM-Symbolic-Plus-2 (GSM-P2), respectively[2].

As shown in Fig. 6, the trend of the evolution of the performance distribution is very consistent across all models: as the difficulty increases, the performance decreases and the variance increases. Note that overall, the *rate of accuracy drop* also increases as the difficulty increases. This is in line with the hypothesis that models are not performing formal reasoning, as the number of required reasoning steps increases linearly, but the rate of drop seems to be faster. Moreover, considering the pattern-matching hypothesis, the increase in variance suggests that searching and pattern-matching become significantly harder for models as the difficulty increases.

[2]Note that adding or removing a clause does not necessarily correspond to increasing or decreasing the number of required reasoning steps by exactly one. However, our main focus in this section is to understand the *evolution* of the performance distribution rather than the precise performance numbers.

GSM-NoOp

```
Oliver picks 44 kiwis on Friday.  Then he picks 58 kiwis on Saturday.  On Sunday, he picks double
the number of kiwis he did on Friday, but five of them were a bit smaller than average.  How many
kiwis does Oliver have?
```

```
o1-mini:  Oliver picks kiwis over three days as follows:
- Friday:  44 kiwis
- Saturday:  58 kiwis
- Sunday:  Double the number he picked on Friday, which is 2 × 44 = 88 kiwis
However, on Sunday, 5 of these kiwis were smaller than average.  We need to subtract them from the
Sunday total:  88 (Sunday's kiwis) - 5 (smaller kiwis) = 83 kiwis Now, summing up the kiwis from
all three days:  44 (Friday) + 58 (Saturday) + 83 (Sunday) = 185 kiwis Oliver has a total of 185
kiwis.
```

```
Llama3-8B:  Let's break it down step by step:
Oliver picks 44 kiwis on Friday.  He picks 58 kiwis on Saturday.  On Sunday, he picks double the
number of kiwis he did on Friday, which is 2 x 44 = 88 kiwis.
Five of the kiwis picked on Sunday are a bit smaller than average, so we subtract 5 from the total
number of kiwis picked on Sunday:  88 - 5 = 83 kiwis.  Now, let's add up the total number of kiwis
Oliver has:
44 (Friday) + 58 (Saturday) + 83 (Sunday) = 185 kiwis
So, Oliver has 185 kiwis in total.
```

Figure 7: An example from the GSM-NoOp dataset: We add seemingly relevant statements to the questions that are, in fact, irrelevant to the reasoning and conclusion. However, the majority of models fail to ignore these statements and blindly convert them into operations, leading to mistakes.

4.4 Can LLMs Really Understand Mathematical Concepts?

In the previous sections, we studied the impact of *type of change* and *difficulty* on the performance distribution. In this section, we demonstrate that models are susceptible to catastrophic performance drops on instances not part of the training distribution, potentially due to their reliance on in-distribution pattern-matching. We introduce GSM-NoOp, a dataset designed to challenge the reasoning capabilities of language models. To create the templates, we add seemingly relevant but ultimately inconsequential statements to GSM-Symbolic templates. Since these statements carry no operational significance, we refer to them as *"No-Op"*.

Fig. 7 illustrates an example from GSM-NoOp. An interesting observation is that models tend to blindly subtract the number of smaller fruits, potentially because their training datasets included similar examples that required conversion to subtraction operations. In the Appendix, we include additional failure cases from GSM-NoOp. Overall, we find that models tend to convert statements to operations without truly understanding their meaning. For instance, a common case we observe is that models interpret statements about "discount" as "multiplication", regardless of the context. This raises the question of whether these models have truly understood the mathematical concepts well enough. Consequently, as shown in Fig. 8a, there is a catastrophic performance decline across all tested models, with the Phi-3-mini model experiencing over a 65% drop, and even stronger models such as o1-preview showing significant declines.

To better understand this performance drop, we conducted another experiment. While our previous evaluations on GSM-P2 used the original 8-shots of GSM8K, here we explore two new scenarios where we change the source of the 8-shots. We report the results in Figures 8b and 8c.

NoOp-Symb (Using GSM-Symbolic shots of the *same* question): During evaluation, we include 8 different shots of the *same* question coming from GSM-Symbolic. Hence, each shot provides the required reasoning steps. The target question from GSM-NoOp then presents yet another variation of the same question that is different only in values and the added clause that is inconsequential. This setup should simplify the task by making it clear that the extra information in the target question is irrelevant. However, as shown in Fig. 8b, the performance remains within the standard deviation, even with 8 shots of the same question providing the reasoning chain. Interestingly, Fig. 8c shows that some models can perform significantly better, even though they don't perform nearly as well on GSM8K and GSM-Symbolic. We believe this is a very notable observation.

NoOp-NoOp (Using GSM-NoOp shots of *different* questions): Here, we provide 8 shots chosen randomly from different questions of GSM-NoOp in the context. These questions share the common